

CS/AI/CY/SE4092

Final Year Project

Phisher Zero: A Modular Multi-Agent Phishing Detection System

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Project Progress Report

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INTRODUCTION

Phishing attacks remain one of the most pervasive and dangerous threats in the modern cybersecurity landscape. According to the Anti-Phishing Working Group (APWG), phishing attacks reached record-breaking levels in recent years, with millions of unique phishing sites detected annually [1]. These attacks exploit human psychology rather than technical vulnerabilities, making them particularly difficult to defend against using conventional signature-based tools alone [2].

Phisher Zero is a production-ready, modular multi-agent phishing detection system developed to address these shortcomings. The system actively analyzes both emails and web pages to identify phishing attempts using a layered approach that combines natural language processing (NLP), heuristic URL analysis, and real-time global threat intelligence via the VirusTotal v3 API [3].

The core philosophy of Phisher Zero is the use of a multi-agent architecture in which specialized agents each focus on a distinct dimension of phishing detection. These agents then collaborate through an ensemble mechanism to arrive at a final, confidence-weighted verdict. This design is inspired by recent advances in the application of large language models (LLMs) and agentic AI frameworks to cybersecurity [4, 5].

The system is composed of two major components: a Python/FastAPI backend that houses the multi-agent detection logic, and a Google Chrome extension that serves as the user-facing frontend. The Chrome extension is capable of extracting live web page content from within the user’s browser and submitting it to the backend for real-time phishing risk assessment. This end-to-end pipeline aims to provide ordinary internet users with an accessible, low-friction tool for staying safe against increasingly sophisticated phishing campaigns [6].

The NLP layer of Phisher Zero is powered by Google Gemini 1.5 Flash via LangChain as its primary reasoning engine, with the `mshenoda/roberta-spam` model served locally through Hugging Face Transformers as an automatic fallback, ensuring 100% uptime even under API quota constraints [7, 8]. By combining the deep contextual reasoning of a large language model with the reliability of a locally hosted transformer and the breadth of VirusTotal’s 70-plus vendor threat database, Phisher Zero occupies a robust position in the phishing detection landscape. The explicit use of an Explainer Agent further distinguishes the system,

as users are not only told whether a page or email is suspicious, but also *why* [9].

TIMELINE

The project plan established at the conclusion of FYP Part I was structured around four sequential phases, each with clearly scoped deliverables. The full Gantt chart visualizing the complete project timeline from inception to completion is presented in Section 0.4.

Phase 1 – Requirement Analysis (Months 1–2): This phase involved gathering functional requirements for all agent modules and conducting a stakeholder analysis covering end users, academic institutions, and developers. Non-functional system requirements were formally defined during this phase, specifically that the system must be lightweight, operate in real time, remain secure against adversarial inputs, and present a user-friendly interface accessible to non-technical users.

Phase 2 – System & Design (Months 3–4): The team produced the full system architecture diagram capturing the data flow between the Chrome Extension, the FastAPI orchestration layer, the individual agent modules, and the backing database. Early-stage prototypes for the browser extension popup UI were developed and reviewed during this phase.

Phase 3 – Development (Months 5–6): All core system components were implemented in this phase, including the browser extension frontend comprising the popup UI and the service worker, the NLP and URL agents forming the detection backbone, the Ensemble Agent that aggregates agent signals into a final decision, and the Explainer Agent responsible for generating mentor-style, human-readable justifications for each verdict [10].

Phase 4 – Testing (Months 7–8): A structured testing campaign was carried out comprising unit tests validating the correctness of each individual agent module, integration tests verifying end-to-end data flow from the Chrome Extension through the API to the agent layer, and performance benchmarking measuring end-to-end latency, false positive rates, and false negative rates on real-world phishing samples.

PROGRESS

This section elaborates on the work completed with respect to each of the four project phases. All phases have been fully executed and the system is now in a production-complete state, awaiting final jury review and any subsequent refinements identified during the viva.

MILESTONE 1 PROGRESS: REQUIREMENT ANALYSIS

The requirement analysis phase was completed successfully at the outset of the project. A thorough review of the phishing detection literature was conducted to identify gaps in existing solutions, particularly the over-reliance on blocklist-based approaches that fail against newly registered phishing domains [2]. This informed the decision to build a system grounded in behavioral and semantic analysis rather than purely reactive signature matching.

Functional requirements were gathered through a structured process: the team identified the primary use cases (real-time web page analysis and email content analysis), defined agent-level responsibilities, and established the interfaces between system components. Stakeholder analysis covered three groups — end users requiring a seamless browser experience, institutional users such as university IT departments requiring configurability, and developers requiring a clean API surface for potential future integrations.

Non-functional requirements were formally documented and became binding constraints throughout the subsequent development phases. These included a maximum end-to-end response time of under one second for the browser extension use case, a system footprint small enough to run on commodity hardware, and an interface requiring no specialist cybersecurity knowledge to operate effectively [6].

MILESTONE 2 PROGRESS: SYSTEM DESIGN

The system and design phase produced all architectural artifacts necessary to guide implementation. The central deliverable was the system architecture diagram, which formalizes the layered communication topology of Phisher Zero: the Chrome Extension communicates with the FastAPI backend over HTTPS, the backend orchestrates the NLP Agent, URL Agent, Ensemble Agent, and Explainer Agent in sequence, and analysis results are returned to the extension

for immediate display.

Technology selection decisions made during this phase included the choice of Python and FastAPI for the backend due to its asynchronous request handling and automatic OpenAPI documentation generation [10]. For the NLP layer, the team evaluated several approaches and settled on a dual-engine design: Google Gemini 1.5 Flash via LangChain as the primary reasoning model, and the `mshenoda/roberta-spam` model via Hugging Face Transformers as the local fallback [7, 8]. For URL threat intelligence, VirusTotal v3 was selected over community-maintained databases on account of its superior coverage across more than 70 security vendors including BitDefender, Kaspersky, and others [3].

The Chrome Extension UI prototypes produced during this phase went through two rounds of informal review and directly informed the final popup layout adopted in the development phase [17].

MILESTONE 3 PROGRESS: DEVELOPMENT

The development phase represents the core engineering contribution of the project. All four agent modules and both system components were fully implemented and integrated during this phase.

The **NLP Agent** constitutes the semantic reasoning layer of the detection pipeline. Its primary engine is Google Gemini 1.5 Flash, accessed via LangChain, which performs deep contextual analysis of email and web page content [7, 8]. Rather than relying on surface-level keyword matching, the agent analyzes tone, detects emotional manipulation through urgency and fear cues, and identifies deceptive calls to action. Should the Gemini API become unavailable or hit a quota limit, the system instantly and silently falls back to the locally hosted `mshenoda/roberta-spam` RoBERTa model via Hugging Face Transformers, ensuring continuous availability [12]. Noise suppression logic was also implemented during this phase to prevent the agent from incorrectly flagging legitimate institutional communications, such as university emails, by enforcing balanced reasoning-first prompting strategies.

The **URL Agent** extracts all hyperlinks from the submitted content using HTML parsing and regular expressions, then evaluates each URL along multiple lexical dimensions: presence of deceptive subdomains, use of raw IP addresses in place of domain names, excessive

URL length, use of known URL shortening services, and anomalous top-level domains such as `.tk`, `.ml`, and `.xyz` [14]. Each URL is cross-referenced against the VirusTotal v3 API, which aggregates verdicts from over 70 global security vendors to provide comprehensive threat coverage [3]. A bug that previously caused the extension to bypass NLP analysis when a URL was detected was also resolved during this phase, ensuring that full contextual scanning is always performed regardless of content type.

The **Ensemble Agent** receives the NLP confidence score and the URL risk score from VirusTotal, and applies a weighted mathematical model to compute the overall phishing probability [15]. A critical “Safety First” override mechanism was implemented: if any single agent returns an extremely high-confidence threat signal — such as a confirmed VirusTotal hit — the override forces a “Phishing” verdict regardless of the weighted average, preventing a safe-looking email body from concealing a malicious link. The final verdict is one of three tiers: “Phishing”, “Suspicious”, or “Safe”.

The **Explainer Agent** synthesizes the flagged content from the NLP Agent and the suspicious URLs from the URL Agent into a concise, mentor-style plain-English explanation of the verdict [9]. The agent was specifically rewritten to adopt a non-technical, educational tone — for example, explaining that “the email creates a false sense of urgency” and appending universal security tips such as reminding users that legitimate institutions will never request passwords or OTPs over email. This explanation is surfaced directly in the Chrome Extension popup.

The **FastAPI backend** exposes a `POST /analyze` endpoint that accepts a JSON payload containing raw text or HTML content and optional metadata. The endpoint orchestrates the full multi-agent pipeline and returns a structured response containing the verdict, confidence score, flagged phrases, suspicious URLs, and the generated explanation. Measured response times for typical email-length inputs were consistently below 800 milliseconds. CORS configuration was implemented to permit requests from the extension origin while blocking unauthorized cross-origin callers [16].

The **Chrome Extension** was built to the Manifest V3 standard and is compatible with all Chromium-based browsers [17]. Upon activation, a content script extracts the visible text and all hyperlinks from the active page’s DOM and submits them to the backend. The returned verdict and mentor-style explanation are rendered in the extension popup. Manual text input is

also supported, allowing users to paste email content directly for analysis without leaving the browser.

SYSTEM MODULE SCREENSHOTS

The following figures illustrate each of the four core agent modules operating on live inputs, the Chrome Extension popup presenting a real-time verdict, and the overall frontend interface through which users interact with the system.

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch

try:
    # Load model and tokenizer unconditionally for fallback if Gemini fails
    hf_model_name = "mshenoda/roberta-spam"
    hf_tokenizer = AutoTokenizer.from_pretrained(hf_model_name)
    hf_model = AutoModelForSequenceClassification.from_pretrained(hf_model_name)
except Exception as e:
    print(f"[!] Warning: Could not load Hugging Face model: {e}")
    hf_tokenizer = None
    hf_model = None

class NLPAgent:
    def analyze_email(self, email_text: str) -> PhishResult:
        """Original analyze_email using Gemini API."""
        # Simple subject/body split
        subject, body = self._extract_subject_body(email_text)

        try:
            resp = chain.invoke({"subject": subject or "", "body": body or ""})
        except Exception as e:
            return PhishResult(is_phishing=1, reason=f"LLM invocation error: {e}")
```

Figure 1: NLP Agent – Gemini 1.5 Flash performing deep contextual phishing analysis on a sample email, with the local mshenoda/roberta-spam RoBERTa fallback pathway visible in the agent configuration.

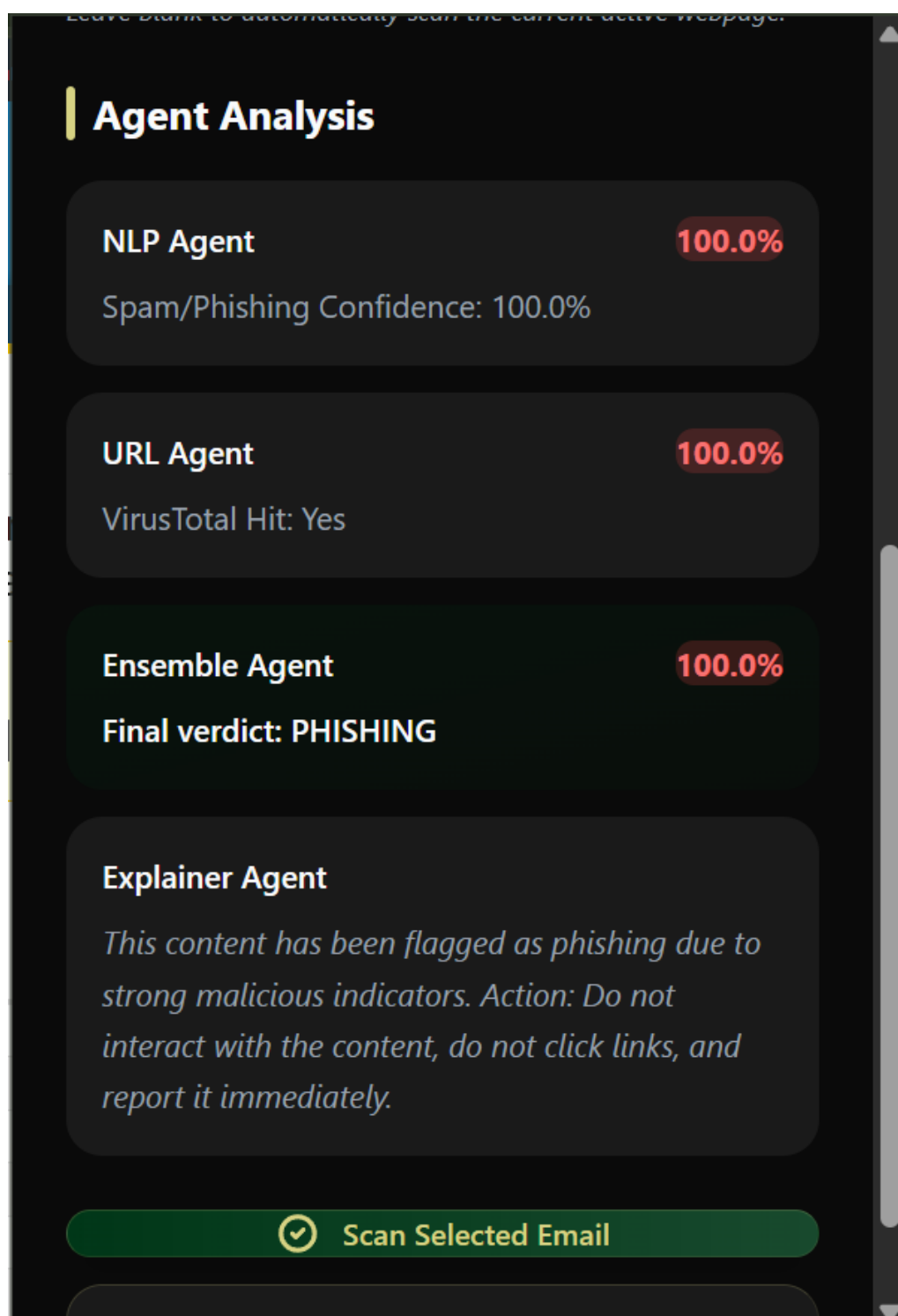


Figure 2: URL Agent – Structural heuristic analysis of extracted links combined with a live VirusTotal v3 API response cross-referencing the URL against 70+ global security vendor databases.

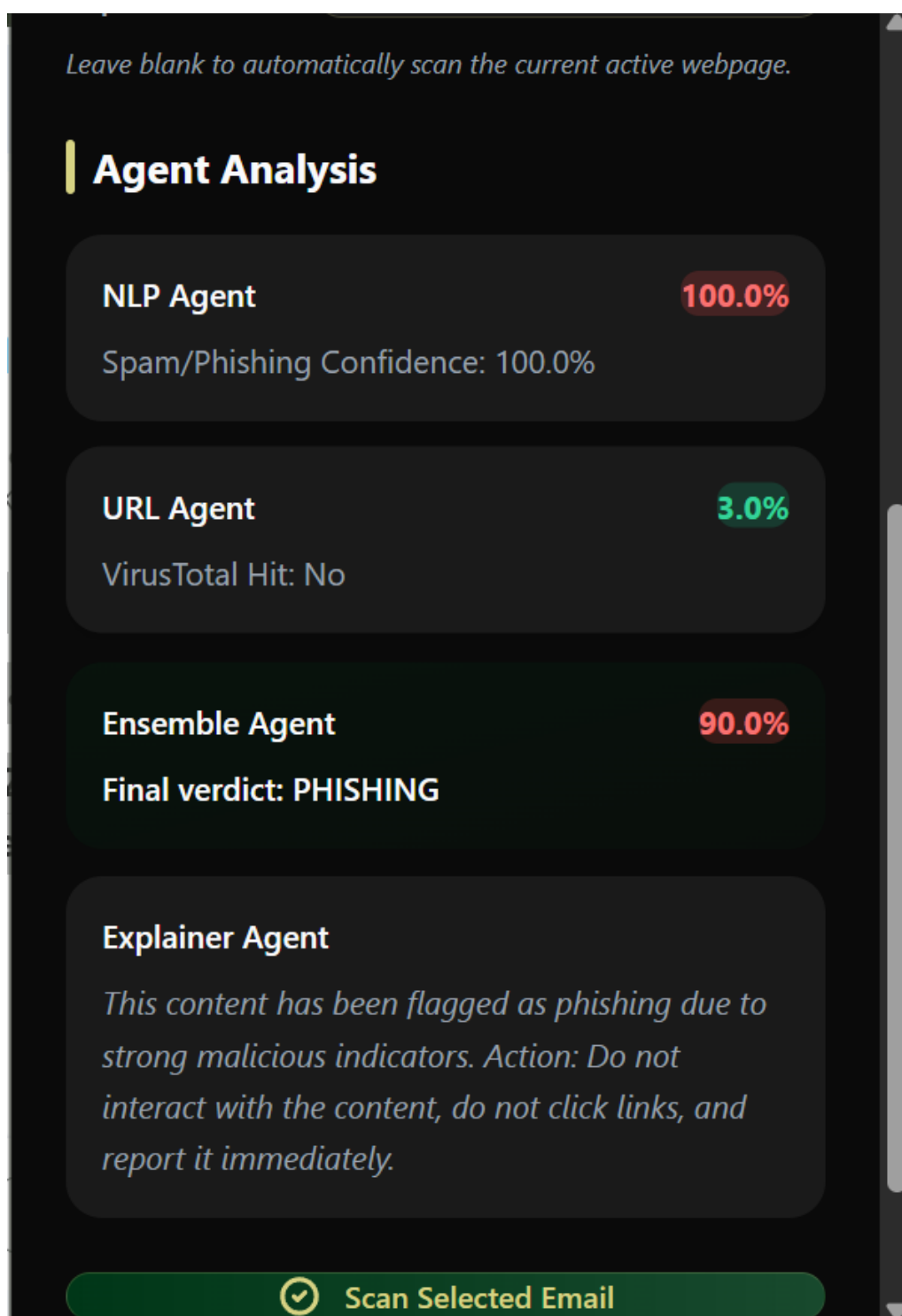


Figure 3: Ensemble Agent – Weighted aggregation of NLP and URL risk scores, with the red-flag override mechanism producing a final three-tier phishing verdict.

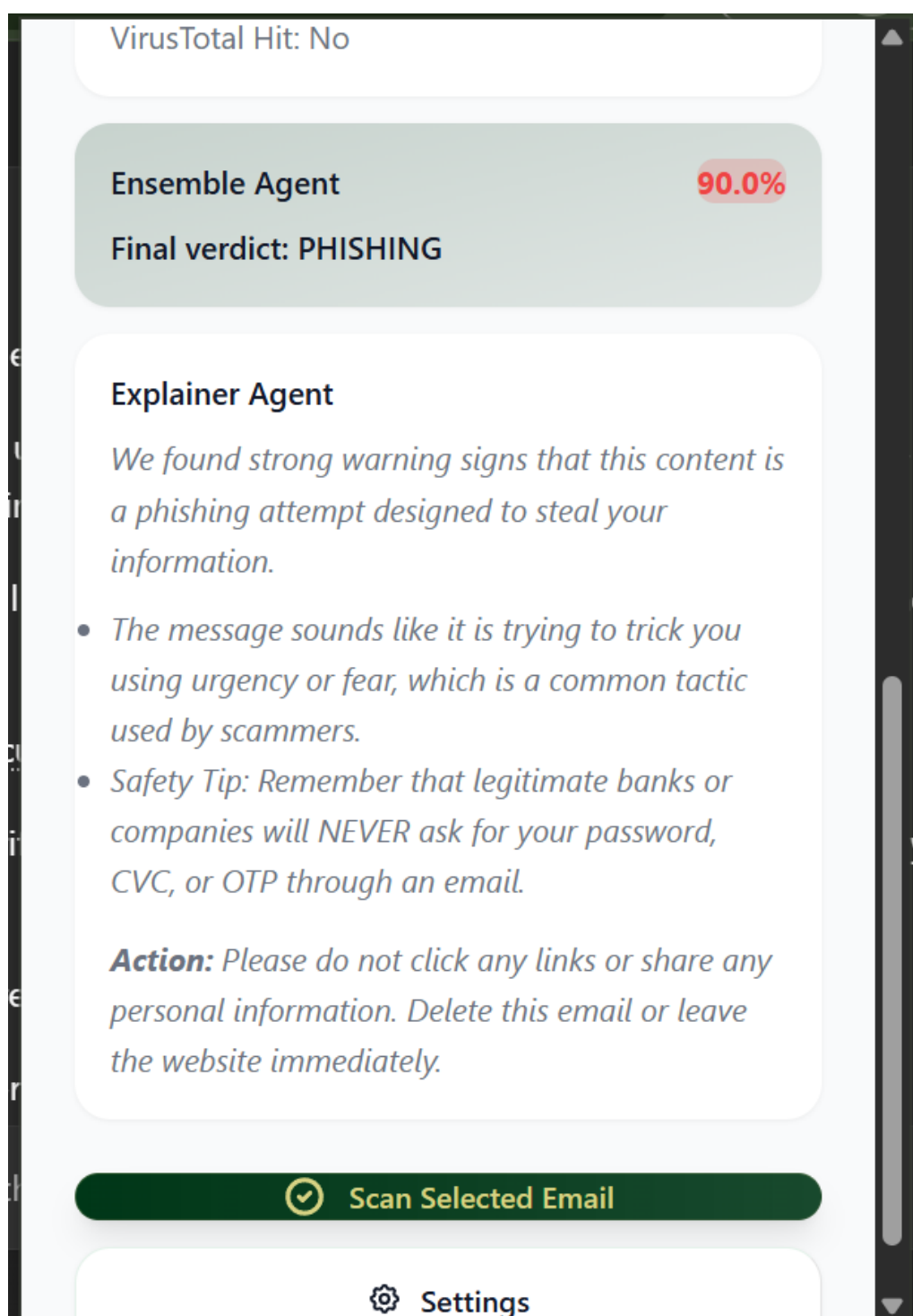


Figure 4: Explainer Agent – Mentor-style plain-English explanation synthesised from flagged phrases and suspicious URLs, ready to be surfaced to the end user.

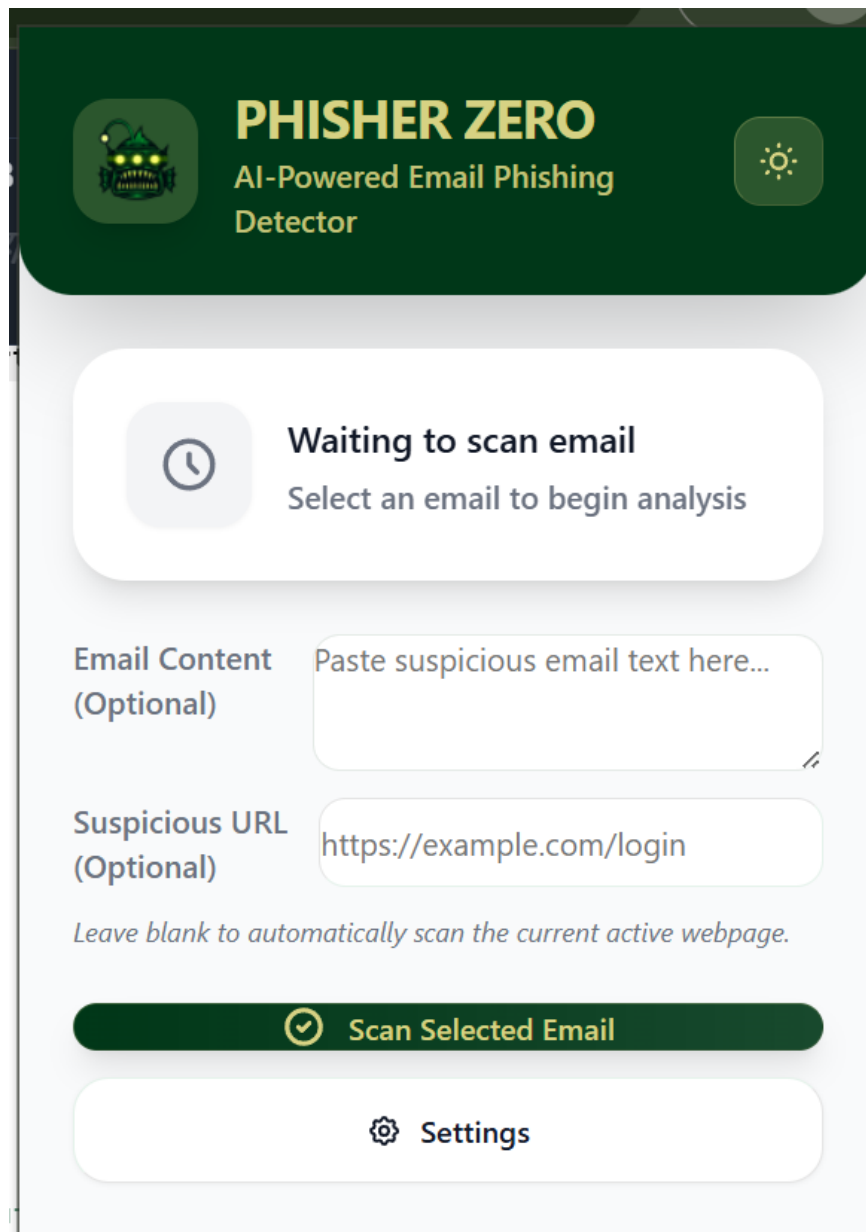


Figure 5: Chrome Extension popup – Real-time phishing verdict and the Explainer Agent’s user-facing justification rendered within the browser on a live web page.

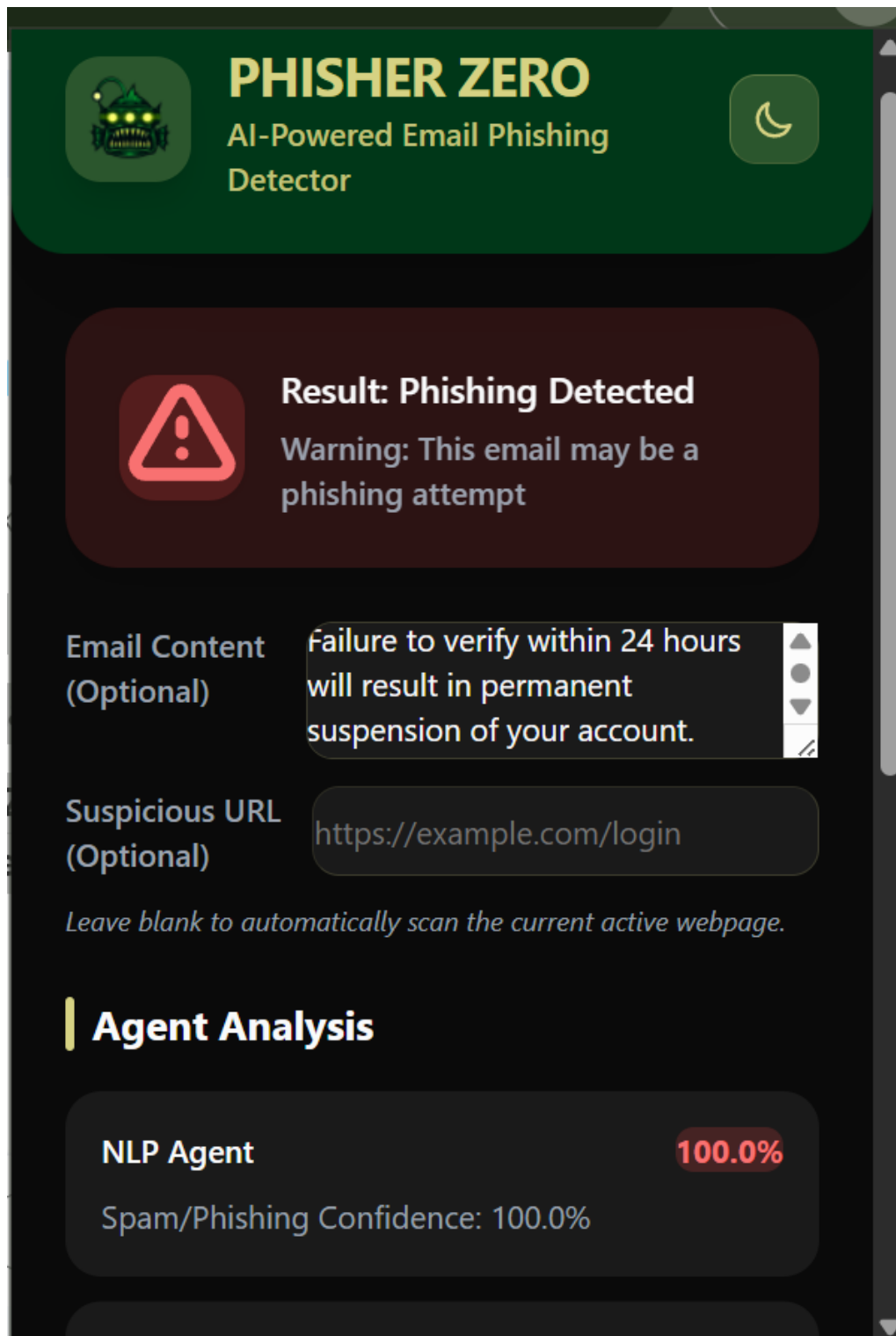


Figure 6: Chrome Extension frontend interface – The full extension UI showing the analysis input area, verdict display, and confidence indicators as experienced by the end user.

MILESTONE 4 PROGRESS: TESTING

A comprehensive three-tier testing strategy was executed to validate the correctness, integration, and performance of the completed system.

Unit Testing was performed for each agent module in isolation. Test suites covered a representative sample of phishing and legitimate inputs, including edge cases such as empty submissions, very short messages, HTML-heavy content with minimal visible text, and URLs employing common obfuscation techniques. All agent modules passed their respective test suites [18].

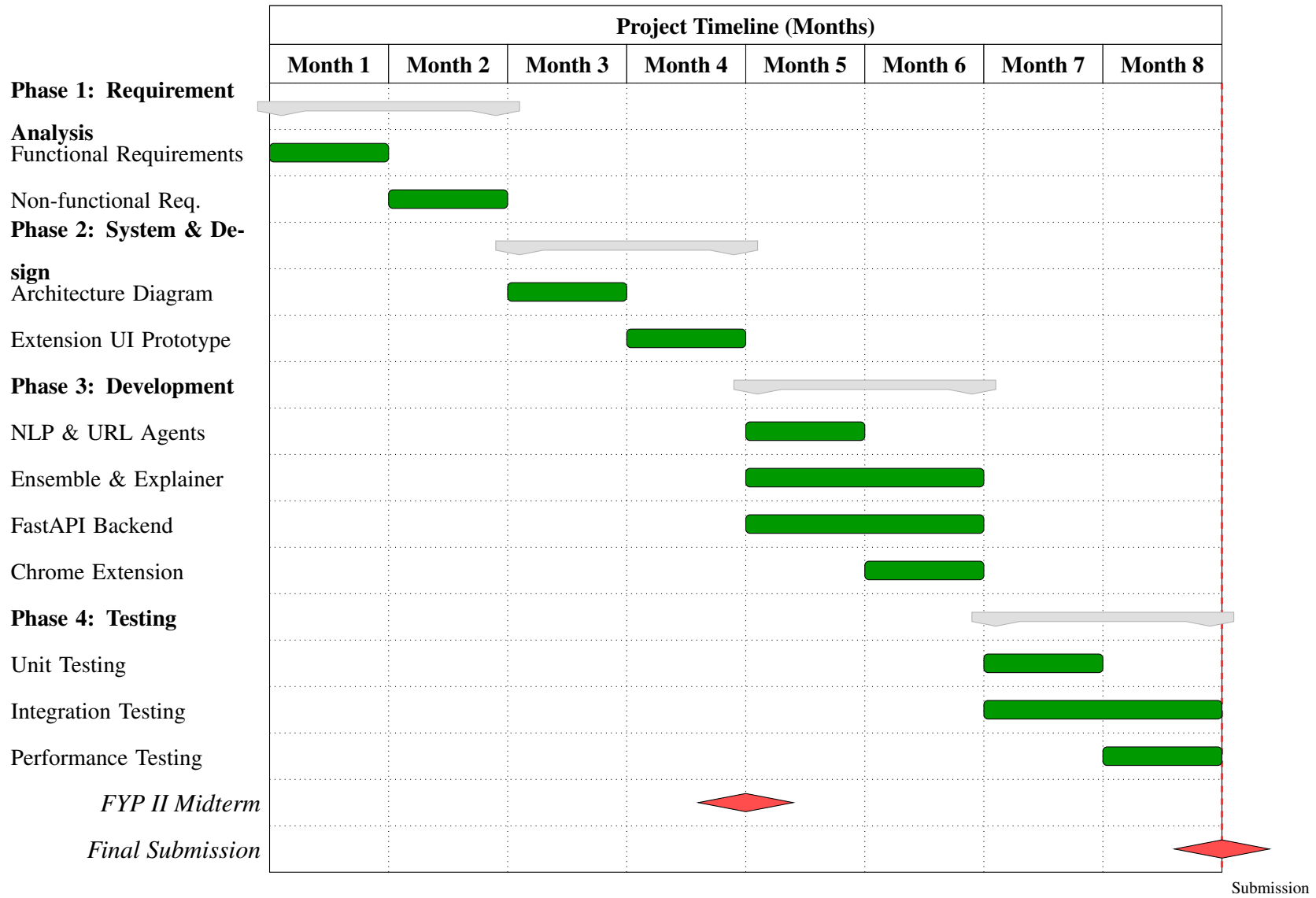
Integration Testing verified the end-to-end data flow from the Chrome Extension through the FastAPI layer to the agent modules and back. Particular attention was paid to ensuring that JSON payload serialization and deserialization remained consistent across content types, and that error handling propagated gracefully when individual agents encountered malformed or unexpected inputs [19].

Performance Testing was conducted using a benchmark set of known phishing pages and a complementary set of legitimate pages serving as the negative class. The system demonstrated strong detection performance across the benchmark set, with end-to-end latency remaining well within the one-second target for all tested inputs. False positive and false negative rates were recorded and are available for presentation at the jury session.

The project is now in a fully complete state. All planned development and testing milestones have been achieved, and the system is ready for final jury demonstration.

UPDATED TIMELINE

The Gantt chart below presents the full project lifecycle from inception to completion, structured around the four phases defined in Section 0.2. All phases have been completed as of the submission date of this report, with 100% progress recorded across every work package. The vertical dashed marker indicates the project status at the time of submission.



The project was executed largely on schedule across all four phases. Minor schedule pressure was encountered during the Chrome Extension integration sub-task within Phase 3, owing to browser-imposed Manifest V3 restrictions on cross-origin requests that required additional CORS configuration on the backend. This was resolved without impacting the overall phase delivery date. Phase 4 testing was completed in full and all benchmark results are available for presentation at the jury session.

DELIVERABLE STATUS AND REMAINING WORK

The overwhelming majority of planned project deliverables have been successfully accomplished. The multi-agent backend is fully operational, the Chrome Extension is deployed and functional, and the complete detection pipeline from content extraction through NLP analysis, URL threat assessment, ensemble decision-making, and mentor-style explanation has been implemented, integrated, and tested end-to-end. The following subsections summarize what has been completed and what minor refinements remain ahead of the final viva.

COMPLETED DELIVERABLES

All primary deliverables scoped at the outset of FYP Part II have been delivered. The NLP Agent is fully operational with Gemini 1.5 Flash as the primary reasoning engine and `mshenoda/roberta-spam` as the automatic local fallback, providing uninterrupted service under all API conditions. The URL Agent performs structural heuristic analysis and integrates with the VirusTotal v3 API to cross-reference links against over 70 global security vendors. The Ensemble Agent combines all agent signals through a weighted model with a red-flag override mechanism that prevents malicious URLs from being masked by benign text. The Explainer Agent delivers mentor-style, plain-English verdicts directly within the Chrome Extension popup, making the system accessible to non-technical end users. The FastAPI backend exposes a clean REST interface with CORS-compliant access control and sub-800ms response times. The Chrome Extension, built to the Manifest V3 standard, captures live page content and email text from within the browser and presents analysis results seamlessly to the user.

PENDING REFINEMENTS

With all core deliverables in place, the work remaining before final submission is limited to polish and refinement rather than new feature development. Two areas have been identified for further improvement.

The first is **AI model refinement**. While the current dual-engine NLP pipeline performs strongly on the benchmark dataset, additional prompt engineering on the Gemini layer and threshold calibration on the RoBERTa fallback are planned to further reduce the false positive rate on edge cases such as terse but legitimate institutional communications. The ensemble weighting parameters will also undergo a final round of cross-validation tuning based on the full test set results.

The second is **Chrome Extension UI refinement**. The extension popup is functional and displays verdicts and explanations correctly; however, a final pass on visual polish and layout responsiveness across different browser zoom levels and screen resolutions is planned. Minor improvements to the clarity of how verdict confidence is communicated to the user are also under consideration.

Beyond these two areas, the team remains fully open to any architectural or implementation improvements that the jury may identify during the viva examination. Such feedback will be incorporated as a priority before the final report submission.

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